Experiment - 2

SQL SCRIPT -

```
CREATE DATABASE a1;
USE a1;

CREATE TABLE student (
   rollno INT,
   name VARCHAR(30)
);

INSERT INTO student VALUES
(1, 'Amit'),
(2, 'Rahul'),
(3, 'Sneha');

CREATE TABLE emp (
   empno INT,
   ename VARCHAR(20) NOT NULL,
   job VARCHAR(10) NOT NULL,
   deptno INT,
   sal DECIMAL(7,2),
   experience INT,
   CONSTRAINT emp_pk PRIMARY KEY (empno),
   CONSTRAINT emp_ck_empno CHECK (empno > 100)
);

ALTER TABLE emp
ADD COLUMN email VARCHAR(50);

ALTER TABLE emp
MODIFY COLUMN job VARCHAR(12);

ALTER TABLE emp
ADD CONSTRAINT emp_ck_experience CHECK (experience >= 0);

ALTER TABLE emp
DROP COLUMN email;

CREATE TABLE dept (
   deptno INT PRIMARY KEY,
   dname VARCHAR(10),
   loc VARCHAR(10)
);

CREATE TABLE employee (
   empno INT,
   ename VARCHAR(25) NOT NULL,
   job VARCHAR(15),
   mgr INT,
```

```
    joindate DATE,
    salary DECIMAL(7,2),
    comm DECIMAL(7,2),
    deptno INT,
    CONSTRAINT emp_fk_dept
        FOREIGN KEY (deptno) REFERENCES dept(deptno)
);

ALTER TABLE employee
ADD CONSTRAINT emp_uk_empno UNIQUE (empno);

ALTER TABLE employee
ALTER COLUMN job SET DEFAULT 'Employee';

ALTER TABLE employee
ADD CONSTRAINT emp_ck_salary CHECK (salary >= 0);

INSERT INTO dept VALUES
(12, 'computer', 'pune'),
(13, 'IT', 'mumbai'),
(14, 'HR', 'delhi');

INSERT INTO employee
(empno, ename, job, mgr, joindate, salary, comm, deptno)
VALUES
(23, 'shiv', 'manager', 55, '2019-09-12', 30000.00, 55.00, 12),
(24, 'rahul', 'developer', 55, '2020-05-15', 35000.00, 100.00, 12),
(25, 'sneha', 'analyst', 56, '2021-03-20', 28000.00, 75.00, 13);

SELECT * FROM student;
SELECT * FROM dept;
SELECT * FROM employee;

INSERT INTO employee
(empno, ename, mgr, joindate, salary, comm, deptno)
VALUES
(26, 'amit', 55, '2022-01-10', 25000.00, 50.00, 14);

INSERT INTO employee
(empno, ename, job, mgr, joindate, salary, comm, deptno)
VALUES
(27, 'test', 'developer', 55, '2023-01-10', -5000.00, 50.00, 12);

TRUNCATE TABLE employee;

DROP TABLE employee;
DROP TABLE dept;
DROP TABLE emp;
DROP TABLE student;
```

OUTPUT -

```
--------------
SELECT * FROM student
--------------

+--------+-------+
| rollno | name  |
+--------+-------+
|      1 | Amit  |
|      2 | Rahul |
|      3 | Sneha |
+--------+-------+
--------------
SELECT * FROM dept
--------------

+--------+----------+--------+
| deptno | dname    | loc    |
+--------+----------+--------+
|     12 | computer | pune   |
|     13 | IT       | mumbai |
|     14 | HR       | delhi  |
+--------+----------+--------+
--------------
SELECT * FROM employee
--------------

+-------+-------+-----------+------+------------+----------+--------+--------+
| empno | ename | job       | mgr  | joindate   | salary   | comm   | deptno |
+-------+-------+-----------+------+------------+----------+--------+--------+
|    23 | shiv  | manager   |   55 | 2019-09-12 | 30000.00 |  55.00 |     12 |
|    24 | rahul | developer |   55 | 2020-05-15 | 35000.00 | 100.00 |     12 |
|    25 | sneha | analyst   |   56 | 2021-03-20 | 28000.00 |  75.00 |     13 |
+-------+-------+-----------+------+------------+----------+--------+--------+
```

Experiment - 3

SQL SCRIPT -

```
CREATE DATABASE IF NOT EXISTS a1;
USE a1;

CREATE TABLE IF NOT EXISTS student (
   rollno INT PRIMARY KEY,
   name VARCHAR(30),
   branch VARCHAR(20),
   marks INT
);

INSERT INTO student (rollno, name, branch, marks) VALUES
(1, 'Amit', 'CSE', 78),
(2, 'Rahul', 'AIML', 85),
(3, 'Sneha', 'CSE', 91),
(4, 'Priya', 'AIML', 72),
(5, 'Rohan', 'CSE', 88);

SHOW DATABASES;

USE a1;

SHOW TABLES;

SELECT * FROM student;

UPDATE student
SET marks = 90
WHERE rollno = 2;

UPDATE student
SET branch = 'AIML', marks = 80
WHERE rollno = 4;

SELECT * FROM student;

DELETE FROM student
WHERE rollno = 5;

SELECT * FROM student;
```

OUTPUT -

```
--------------
SHOW DATABASES
--------------

+--------------------+
| Database           |
+--------------------+
| a1                 |
| bank_management    |
| information_schema |
| join_management    |
| mysql              |
| performance_schema |
| product_management |
| sys                |
+--------------------+
```

```
--------------
SHOW TABLES
--------------

+--------------+
| Tables_in_a1 |
+--------------+
| student      |
+--------------+
```

```
--------------
SELECT * FROM student
--------------

+--------+-------+--------+-------+
| rollno | name  | branch | marks |
+--------+-------+--------+-------+
|      1 | Amit  | CSE    |    78 |
|      2 | Rahul | AIML   |    85 |
|      3 | Sneha | CSE    |    91 |
|      4 | Priya | AIML   |    72 |
|      5 | Rohan | CSE    |    88 |
+--------+-------+--------+-------+
```

```
--------------
SELECT * FROM student
--------------

+--------+-------+--------+-------+
| rollno | name  | branch | marks |
+--------+-------+--------+-------+
|      1 | Amit  | CSE    |    78 |
|      2 | Rahul | AIML   |    90 |
|      3 | Sneha | CSE    |    91 |
|      4 | Priya | AIML   |    80 |
|      5 | Rohan | CSE    |    88 |
+--------+-------+--------+-------+
```

```
--------------
SELECT * FROM student
--------------

+--------+-------+--------+-------+
| rollno | name  | branch | marks |
+--------+-------+--------+-------+
|      1 | Amit  | CSE    |    78 |
|      2 | Rahul | AIML   |    90 |
|      3 | Sneha | CSE    |    91 |
|      4 | Priya | AIML   |    80 |
+--------+-------+--------+-------+
nitesh@nitesh-Latitude-7480:~/Downloads$
```

Experiment - 4

SQL SCRIPT -

```
CREATE DATABASE IF NOT EXISTS a1;
USE a1;

CREATE TABLE employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   department VARCHAR(30),
   salary DECIMAL(10,2)
);

INSERT INTO employee VALUES
(1, 'Amit', 'CSE', 45000),
(2, 'Rahul', 'AIML', 50000),
(3, 'Sneha', 'CSE', 55000),
(4, 'Priya', 'HR', 40000);

SELECT * FROM employee;

ALTER TABLE employee
ADD COLUMN email VARCHAR(100);

ALTER TABLE employee
MODIFY COLUMN emp_name VARCHAR(60);

ALTER TABLE employee
DROP COLUMN email;

SELECT * FROM employee;

RENAME TABLE employee TO staff;

SELECT * FROM staff;

TRUNCATE TABLE staff;

SELECT * FROM staff;

INSERT INTO staff VALUES
(1, 'Amit', 'CSE', 45000),
(2, 'Rahul', 'AIML', 50000),
(3, 'Sneha', 'CSE', 55000);

SELECT * FROM staff;

RENAME TABLE staff TO employee;

SELECT * FROM employee;
```

DROP TABLE employee;

SHOW TABLES;

Backup Command: mysqldump -u root -p a1 > a1_backup.sql
Restore Command: mysql -u root -p a1 < a1_backup.sql

OUTPUT -

```
--------------
SHOW TABLES
--------------

+--------------+
| Tables_in_a1 |
+--------------+
| employee     |
+--------------+
```

```
--------------
SELECT * FROM employee
--------------

+--------+----------+------------+----------+
| emp_id | emp_name | department | salary   |
+--------+----------+------------+----------+
|      1 | Amit     | CSE        | 45000.00 |
|      2 | Rahul    | AIML       | 50000.00 |
|      3 | Sneha    | CSE        | 55000.00 |
|      4 | Priya    | HR         | 40000.00 |
+--------+----------+------------+----------+
```

```
--------------
SELECT * FROM employee
--------------

+--------+----------+------------+----------+-------+
| emp_id | emp_name | department | salary   | email |
+--------+----------+------------+----------+-------+
|      1 | Amit     | CSE        | 45000.00 | NULL  |
|      2 | Rahul    | AIML       | 50000.00 | NULL  |
|      3 | Sneha    | CSE        | 55000.00 | NULL  |
|      4 | Priya    | HR         | 40000.00 | NULL  |
+--------+----------+------------+----------+-------+
```

```
--------------
RENAME TABLE employee TO staff
--------------


--------------
SELECT * FROM staff
--------------

+--------+----------+------------+----------+
| emp_id | emp_name | department | salary   |
+--------+----------+------------+----------+
|      1 | Amit     | CSE        | 45000.00 |
|      2 | Rahul    | AIML       | 50000.00 |
|      3 | Sneha    | CSE        | 55000.00 |
|      4 | Priya    | HR         | 40000.00 |
+--------+----------+------------+----------+
```

```
--------------
SHOW TABLES
--------------

+--------------+
| Tables_in_a1 |
+--------------+
| staff        |
+--------------+
```

Experiment - 5

SQL SCRIPT -

```
CREATE DATABASE a1;
USE a1;

CREATE TABLE product (
   id INT PRIMARY KEY,
   pname VARCHAR(50),
   pprice DECIMAL(8,2),
   discount INT,
   created DATE
);

INSERT INTO product VALUES
(1, 'AMIT', 2000, 100, '2025-07-01'),
(2, 'CORNFLAKES', 800, 50, '2025-06-15'),
(3, 'PARLE', 90, 5, '2025-05-20'),
(4, 'NESTLE', 70, 2, '2025-04-10'),
(5, 'CADBURY', 100, 10, '2025-03-05');

SELECT * FROM product;

SELECT
   COUNT(id) AS numProducts,
   SUM(pprice) AS totalPrice,
   AVG(pprice) AS avgPrice,
   MAX(discount) AS maxDisc,
   MIN(pprice) AS minPrice
FROM product;

SELECT
   discount,
   COUNT(*) AS cnt,
   AVG(pprice) AS avgPrice
FROM product
GROUP BY discount
HAVING COUNT(*) > 1;

SELECT
   DAY(created) AS day,
   MONTH(created) AS month,
   YEAR(created) AS year,
   DATEDIFF(CURDATE(), created) AS daysOld
FROM product;

SELECT
   UPPER(pname) AS upperName,
   LOWER(pname) AS lowerName,
   CONCAT(pname, '_2025') AS tag,
```

```
    SUBSTR(pname, 1, 3) AS prefix,
    REPLACE(pname, 'PAR', 'KIT') AS replacedName
FROM product;

SELECT
    ROUND(pprice * 0.9, 2) AS discountedPrice,
    POWER(pprice, 2) AS priceSquared,
    FLOOR(pprice / 100) * 100 AS floorHundred
FROM product;

SELECT pname
FROM product
WHERE discount > 50
UNION
SELECT pname
FROM product
WHERE pprice < 100;

SELECT pname
FROM product
WHERE discount >= 5
INTERSECT
SELECT pname
FROM product
WHERE pprice < 100;

SELECT pname
FROM product
WHERE discount >= 5
EXCEPT
SELECT pname
FROM product
WHERE pprice < 100;

SELECT
    ROUND(AVG(pprice), 2) AS averagePrice,
    ROUND(AVG(discount), 2) AS averageDiscount
FROM product;
```

OUTPUT -

```
--------------
SELECT * FROM product
--------------

+----+------------+---------+----------+------------+
| id | pname      | pprice  | discount | created    |
+----+------------+---------+----------+------------+
|  1 | AMIT       | 2000.00 |      100 | 2025-07-01 |
|  2 | CORNFLAKES |  800.00 |       50 | 2025-06-15 |
|  3 | PARLE      |   90.00 |        5 | 2025-05-20 |
|  4 | NESTLE     |   70.00 |        2 | 2025-04-10 |
|  5 | CADBURY    |  100.00 |       10 | 2025-03-05 |
+----+------------+---------+----------+------------+
```

```
+-------------+------------+------------+---------+----------+
| numProducts | totalPrice | avgPrice   | maxDisc | minPrice |
+-------------+------------+------------+---------+----------+
|           5 |    3060.00 | 612.000000 |     100 |    70.00 |
+-------------+------------+------------+---------+----------+
```

```
+------+-------+------+---------+
| day  | month | year | daysOld |
+------+-------+------+---------+
|    1 |     7 | 2025 |     459 |
|   15 |     6 | 2025 |     475 |
|   20 |     5 | 2025 |     501 |
|   10 |     4 | 2025 |     541 |
|    5 |     3 | 2025 |     577 |
+------+-------+------+---------+
```

```
+------------+------------+----------------+--------+--------------+
| upperName  | lowerName  | tag            | prefix | replacedName |
+------------+------------+----------------+--------+--------------+
| AMIT       | amit       | AMIT_2025      | AMI    | AMIT         |
| CORNFLAKES | cornflakes | CORNFLAKES_2025 | COR    | CORNFLAKES   |
| PARLE      | parle      | PARLE_2025     | PAR    | KITLE        |
| NESTLE     | nestle     | NESTLE_2025    | NES    | NESTLE       |
| CADBURY    | cadbury    | CADBURY_2025   | CAD    | CADBURY      |
+------------+------------+----------------+--------+--------------+
```

```
+-----------------+-------------+-------------+
| discountedPrice | priceSquared | floorHundred |
+-----------------+-------------+-------------+
|         1800.00 |      4000000 |         2000 |
|          720.00 |       640000 |          800 |
|           81.00 |         8100 |            0 |
|           63.00 |         4900 |            0 |
|           90.00 |        10000 |          100 |
+-----------------+-------------+-------------+
```

```
--------------
SELECT pname
FROM product
WHERE discount > 50
UNION
SELECT pname
FROM product
WHERE pprice < 100
--------------

+--------+
| pname  |
+--------+
| AMIT   |
| PARLE  |
| NESTLE |
+--------+
```

```
--------------
SELECT pname
FROM product
WHERE discount >= 5
INTERSECT
SELECT pname
FROM product
WHERE pprice < 100
--------------

+-------+
| pname |
+-------+
| PARLE |
+-------+
```

```
--------------
SELECT pname
FROM product
WHERE discount >= 5
EXCEPT
SELECT pname
FROM product
WHERE pprice < 100
--------------

+------------+
| pname      |
+------------+
| AMIT       |
| CORNFLAKES |
| CADBURY    |
+------------+
```

```
--------------
SELECT
    ROUND(AVG(pprice), 2) AS averagePrice,
    ROUND(AVG(discount), 2) AS averageDiscount
FROM product
--------------

+--------------+-----------------+
| averagePrice | averageDiscount |
+--------------+-----------------+
|       612.00 |           33.40 |
+--------------+-----------------+
nitesh@nitesh-Latitude-7480:~$
```

Experiment - 6

SQL SCRIPT -

```
CREATE DATABASE a1;
USE a1;

CREATE TABLE dept (
   deptno INT PRIMARY KEY,
   dname VARCHAR(30),
   loc VARCHAR(30)
);

CREATE TABLE emp (
   empno INT PRIMARY KEY,
   ename VARCHAR(30),
   job VARCHAR(30),
   mgr INT,
   salary DECIMAL(10,2),
   dept INT,
   FOREIGN KEY (dept) REFERENCES dept(deptno)
);

CREATE TABLE salgrade (
   grade INT PRIMARY KEY,
   losal DECIMAL(10,2),
   hisal DECIMAL(10,2)
);

INSERT INTO dept VALUES
(10, 'Accounts', 'Pune'),
(20, 'Research', 'Mumbai'),
(30, 'Sales', 'Delhi'),
(40, 'IT', 'Bangalore');

INSERT INTO emp VALUES
(101, 'Scott', 'Manager', NULL, 80000, 10),
(102, 'Amit', 'Developer', 101, 60000, 40),
(103, 'Rahul', 'Developer', 101, 55000, 40),
(104, 'Sneha', 'Analyst', 101, 45000, 20),
(105, 'Priya', 'Sales Executive', 101, 35000, 30),
(106, 'Rohan', 'Intern', 102, 20000, 40);

INSERT INTO salgrade VALUES
(1, 10000, 29999),
(2, 30000, 49999),
(3, 50000, 69999),
(4, 70000, 99999);

CREATE TABLE customers (
   custID INT PRIMARY KEY,
```

```
    cust_name VARCHAR(50),
    city VARCHAR(30)
);

CREATE TABLE orders (
    orderID INT PRIMARY KEY,
    custID INT,
    order_date DATE,
    amount DECIMAL(10,2),
    FOREIGN KEY (custID) REFERENCES customers(custID)
);

INSERT INTO customers VALUES
(1, 'Customer A', 'Pune'),
(2, 'Customer B', 'Mumbai'),
(3, 'Customer C', 'Delhi'),
(4, 'Customer D', 'Nashik');

INSERT INTO orders VALUES
(101, 1, '2026-01-10', 5000),
(102, 2, '2026-01-15', 7500),
(103, 1, '2026-02-05', 3000),
(104, 3, '2026-02-20', 9000);

CREATE TABLE department (
    DNO INT PRIMARY KEY,
    DNAME VARCHAR(50),
    MGRSSN INT
);

CREATE TABLE employee (
    SSN INT PRIMARY KEY,
    NAME VARCHAR(50),
    DNO INT,
    SALARY DECIMAL(10,2),
    FOREIGN KEY (DNO) REFERENCES department(DNO)
);

CREATE TABLE project (
    PNO INT PRIMARY KEY,
    PNAME VARCHAR(50),
    DNO INT,
    FOREIGN KEY (DNO) REFERENCES department(DNO)
);

CREATE TABLE works_on (
    SSN INT,
    PNO INT,
    HOURS DECIMAL(5,2),
    PRIMARY KEY (SSN, PNO),
    FOREIGN KEY (SSN) REFERENCES employee(SSN),
```

```
    FOREIGN KEY (PNO) REFERENCES project(PNO)
);

INSERT INTO department VALUES
(10, 'Accounts', 1001),
(20, 'Research', 1004),
(30, 'Sales', 1005),
(40, 'IT', 1002);

INSERT INTO employee VALUES
(1001, 'Scott', 10, 80000),
(1002, 'Amit', 40, 60000),
(1003, 'Rahul', 40, 55000),
(1004, 'Sneha', 20, 45000),
(1005, 'Priya', 30, 35000);

INSERT INTO project VALUES
(1, 'IoT', 40),
(2, 'Banking System', 10),
(3, 'Research Portal', 20),
(4, 'Sales Analytics', 30);

INSERT INTO works_on VALUES
(1002, 1, 20),
(1003, 1, 15),
(1001, 2, 10),
(1004, 3, 25),
(1005, 4, 30);

SELECT
    e.empno,
    e.ename,
    e.dept,
    d.deptno,
    d.loc,
    d.dname
FROM emp e
INNER JOIN dept d
ON e.dept = d.deptno;

SELECT
    e.ename,
    e.salary,
    s.grade
FROM emp e
INNER JOIN salgrade s
ON e.salary >= s.losal
AND e.salary <= s.hisal;

SELECT
    worker.ename AS employee,
```

```
    manager.ename AS manager
FROM emp worker
INNER JOIN emp manager
ON worker.mgr = manager.empno;

SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
LEFT JOIN customers c
ON o.custID = c.custID;

SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
RIGHT JOIN customers c
ON o.custID = c.custID;

SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
LEFT JOIN customers c
ON o.custID = c.custID
UNION
SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
RIGHT JOIN customers c
ON o.custID = c.custID;

SELECT DISTINCT p.PNO
FROM project p
JOIN department d
ON p.DNO = d.DNO
JOIN employee e
ON d.MGRSSN = e.SSN
WHERE e.NAME = 'Scott'
UNION
SELECT DISTINCT w.PNO
FROM project p
JOIN works_on w
ON p.PNO = w.PNO
JOIN employee e
ON w.SSN = e.SSN
```

```
WHERE e.NAME = 'Scott';

SELECT
   e.NAME,
   1.1 * e.SALARY AS INCREASED_SALARY
FROM employee e
JOIN works_on w
ON e.SSN = w.SSN
JOIN project p
ON w.PNO = p.PNO
WHERE p.PNAME = 'IoT';

SELECT
   SUM(e.SALARY) AS TOTAL_SALARY,
   MAX(e.SALARY) AS MAX_SALARY,
   MIN(e.SALARY) AS MIN_SALARY,
   AVG(e.SALARY) AS AVG_SALARY
FROM employee e
JOIN department d
ON e.DNO = d.DNO
WHERE d.DNAME = 'Accounts';

SELECT
   e.ename,
   d.dname,
   e.salary
FROM emp e
JOIN dept d
ON e.dept = d.deptno;

SELECT
   d.dname,
   COUNT(e.empno) AS employee_count,
   AVG(e.salary) AS average_salary
FROM emp e
RIGHT JOIN dept d
ON e.dept = d.deptno
GROUP BY d.deptno, d.dname;
```

OUTPUT -

```
--------------
SELECT
    e.empno,
    e.ename,
    e.dept,
    d.deptno,
    d.loc,
    d.dname
FROM emp e
INNER JOIN dept d
ON e.dept = d.deptno
--------------
```

empno	ename	dept	deptno	loc	dname
101	Scott	10	10	Pune	Accounts
104	Sneha	20	20	Mumbai	Research
105	Priya	30	30	Delhi	Sales
102	Amit	40	40	Bangalore	IT
103	Rahul	40	40	Bangalore	IT
106	Rohan	40	40	Bangalore	IT

```
--------------
SELECT
    e.ename,
    e.salary,
    s.grade
FROM emp e
INNER JOIN salgrade s
ON e.salary >= s.losal
AND e.salary <= s.hisal
--------------
```

ename	salary	grade
Scott	80000.00	4
Amit	60000.00	3
Rahul	55000.00	3
Sneha	45000.00	2
Priya	35000.00	2
Rohan	20000.00	1

```
--------------
SELECT
    worker.ename AS employee,
    manager.ename AS manager
FROM emp worker
INNER JOIN emp manager
ON worker.mgr = manager.empno
--------------
```

employee	manager
Amit	Scott
Rahul	Scott
Sneha	Scott
Priya	Scott
Rohan	Amit

```
--------------
SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
LEFT JOIN customers c
ON o.custID = c.custID
--------------

+---------+-----------+--------+
| orderID | cust_name  | city   |
+---------+-----------+--------+
|     101 | Customer A | Pune   |
|     103 | Customer A | Pune   |
|     102 | Customer B | Mumbai |
|     104 | Customer C | Delhi  |
+---------+-----------+--------+
```

```
--------------
SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
RIGHT JOIN customers c
ON o.custID = c.custID
--------------

+---------+-----------+--------+
| orderID | cust_name  | city   |
+---------+-----------+--------+
|     101 | Customer A | Pune   |
|     103 | Customer A | Pune   |
|     102 | Customer B | Mumbai |
|     104 | Customer C | Delhi  |
|    NULL | Customer D | Nashik |
+---------+-----------+--------+
```

```
--------------
SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
LEFT JOIN customers c
ON o.custID = c.custID
UNION
SELECT
    o.orderID,
    c.cust_name,
    c.city
FROM orders o
RIGHT JOIN customers c
ON o.custID = c.custID
--------------

+---------+-----------+--------+
| orderID | cust_name  | city   |
+---------+-----------+--------+
|     101 | Customer A | Pune   |
|     103 | Customer A | Pune   |
|     102 | Customer B | Mumbai |
|     104 | Customer C | Delhi  |
|    NULL | Customer D | Nashik |
+---------+-----------+--------+
```

```
--------------
SELECT
    e.NAME,
    1.1 * e.SALARY AS INCREASED_SALARY
FROM employee e
JOIN works_on w
ON e.SSN = w.SSN
JOIN project p
ON w.PNO = p.PNO
WHERE p.PNAME = 'IoT'
--------------

+-------+------------------+
| NAME  | INCREASED_SALARY |
+-------+------------------+
| Amit  |        66000.000 |
| Rahul |        60500.000 |
+-------+------------------+
```

```
--------------
SELECT
    SUM(e.SALARY) AS TOTAL_SALARY,
    MAX(e.SALARY) AS MAX_SALARY,
    MIN(e.SALARY) AS MIN_SALARY,
    AVG(e.SALARY) AS AVG_SALARY
FROM employee e
JOIN department d
ON e.DNO = d.DNO
WHERE d.DNAME = 'Accounts'
--------------

+--------------+------------+------------+--------------+
| TOTAL_SALARY | MAX_SALARY | MIN_SALARY | AVG_SALARY   |
+--------------+------------+------------+--------------+
|     80000.00 |   80000.00 |   80000.00 | 80000.000000 |
+--------------+------------+------------+--------------+
```

```
--------------
SELECT
    e.ename,
    d.dname,
    e.salary
FROM emp e
JOIN dept d
ON e.dept = d.deptno
--------------

+-------+----------+----------+
| ename | dname    | salary   |
+-------+----------+----------+
| Scott | Accounts | 80000.00 |
| Sneha | Research | 45000.00 |
| Priya | Sales    | 35000.00 |
| Amit  | IT       | 60000.00 |
| Rahul | IT       | 55000.00 |
| Rohan | IT       | 20000.00 |
+-------+----------+----------+
```

```
--------------
SELECT
    d.dname,
    COUNT(e.empno) AS employee_count,
    AVG(e.salary) AS average_salary
FROM emp e
RIGHT JOIN dept d
ON e.dept = d.deptno
GROUP BY d.deptno, d.dname
--------------

+----------+----------------+----------------+
| dname    | employee_count | average_salary |
+----------+----------------+----------------+
| Accounts |              1 |   80000.000000 |
| Research |              1 |   45000.000000 |
| Sales    |              1 |   35000.000000 |
| IT       |              3 |   45000.000000 |
+----------+----------------+----------------+
nitesh@nitesh-Latitude-7480:~$
```

Experiment - 7

SQL SCRIPT -

```
CREATE DATABASE a1;
USE a1;

CREATE TABLE departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(50),
    location VARCHAR(50)
);

CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(50),
    salary DECIMAL(10,2),
    department_id INT,
    FOREIGN KEY (department_id) REFERENCES departments(department_id)
);

INSERT INTO departments VALUES
(1, 'IT', 'Pune'),
(2, 'HR', 'New York'),
(3, 'Finance', 'Mumbai'),
(4, 'Sales', 'New York'),
(5, 'Marketing', 'Delhi');

INSERT INTO employees VALUES
(101, 'Amit', 60000, 1),
(102, 'Rahul', 75000, 1),
(103, 'Sneha', 50000, 2),
(104, 'Priya', 65000, 2),
(105, 'Rohan', 45000, 3),
(106, 'Neha', 80000, 3),
(107, 'Karan', 55000, 4),
(108, 'Anjali', 70000, 4),
(109, 'Vikas', 40000, 5),
(110, 'Pooja', 90000, 1);

SELECT * FROM employees
WHERE salary > 60000
AND department_id = 1;

SELECT * FROM employees
WHERE salary >= 50000
AND salary <= 80000;

SELECT * FROM employees
WHERE employee_name LIKE 'A%';
```

```sql
SELECT * FROM employees
WHERE department_id = 1
OR department_id = 2;

SELECT * FROM employees
WHERE NOT department_id = 3;

SELECT
    COUNT(*) AS total_employees,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MAX(salary) AS maximum_salary,
    MIN(salary) AS minimum_salary
FROM employees;

SELECT
    department_id,
    COUNT(*) AS employee_count,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MAX(salary) AS maximum_salary,
    MIN(salary) AS minimum_salary
FROM employees
GROUP BY department_id;

SELECT
    department_id,
    COUNT(*) AS employee_count,
    AVG(salary) AS average_salary
FROM employees
GROUP BY department_id
HAVING AVG(salary) > 60000;

SELECT *
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
);

SELECT *
FROM employees
WHERE department_id IN (
    SELECT department_id
    FROM departments
    WHERE location = 'New York'
);

SELECT *
FROM employees e
WHERE EXISTS (
```

```
    SELECT 1
    FROM departments d
    WHERE d.department_id = e.department_id
    AND d.location = 'New York'
);

SELECT *
FROM employees e
WHERE EXISTS (
    SELECT 1
    FROM departments d
    WHERE d.department_id = e.department_id
    AND d.department_name = 'HR'
);

SELECT *
FROM employees
WHERE department_id IN (
    SELECT department_id
    FROM departments
    WHERE department_name IN ('IT', 'Finance')
);

SELECT *
FROM employees e
WHERE EXISTS (
    SELECT 1
    FROM departments d
    WHERE d.department_id = e.department_id
);

SELECT
    e.employee_name,
    e.salary,
    d.department_name,
    d.location
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE e.salary > (
    SELECT AVG(salary)
    FROM employees
);
```

OUTPUT -

```
--------------
SELECT * FROM employees
WHERE salary > 60000
AND department_id = 1
--------------

+-------------+---------------+----------+---------------+
| employee_id | employee_name | salary   | department_id |
+-------------+---------------+----------+---------------+
|         102 | Rahul         | 75000.00 |             1 |
|         110 | Pooja         | 90000.00 |             1 |
+-------------+---------------+----------+---------------+
```

```
--------------
SELECT * FROM employees
WHERE salary >= 50000
AND salary <= 80000
--------------

+-------------+---------------+----------+---------------+
| employee_id | employee_name | salary   | department_id |
+-------------+---------------+----------+---------------+
|         101 | Amit          | 60000.00 |             1 |
|         102 | Rahul         | 75000.00 |             1 |
|         103 | Sneha         | 50000.00 |             2 |
|         104 | Priya         | 65000.00 |             2 |
|         106 | Neha          | 80000.00 |             3 |
|         107 | Karan         | 55000.00 |             4 |
|         108 | Anjali        | 70000.00 |             4 |
+-------------+---------------+----------+---------------+
```

```
--------------
SELECT * FROM employees
WHERE employee_name LIKE 'A%'
--------------

+-------------+---------------+----------+---------------+
| employee_id | employee_name | salary   | department_id |
+-------------+---------------+----------+---------------+
|         101 | Amit          | 60000.00 |             1 |
|         108 | Anjali        | 70000.00 |             4 |
+-------------+---------------+----------+---------------+
```

```
--------------
SELECT * FROM employees
WHERE department_id = 1
OR department_id = 2
--------------
```

```
+-------------+---------------+----------+---------------+
| employee_id | employee_name | salary   | department_id |
+-------------+---------------+----------+---------------+
|         101 | Amit          | 60000.00 |             1 |
|         102 | Rahul         | 75000.00 |             1 |
|         110 | Pooja         | 90000.00 |             1 |
|         103 | Sneha         | 50000.00 |             2 |
|         104 | Priya         | 65000.00 |             2 |
+-------------+---------------+----------+---------------+
```

```
--------------
SELECT * FROM employees
WHERE NOT department_id = 3
--------------
```

```
+-------------+---------------+----------+---------------+
| employee_id | employee_name | salary   | department_id |
+-------------+---------------+----------+---------------+
|         101 | Amit          | 60000.00 |             1 |
|         102 | Rahul         | 75000.00 |             1 |
|         110 | Pooja         | 90000.00 |             1 |
|         103 | Sneha         | 50000.00 |             2 |
|         104 | Priya         | 65000.00 |             2 |
|         107 | Karan         | 55000.00 |             4 |
|         108 | Anjali        | 70000.00 |             4 |
|         109 | Vikas         | 40000.00 |             5 |
+-------------+---------------+----------+---------------+
```

```
--------------
SELECT
    COUNT(*) AS total_employees,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MAX(salary) AS maximum_salary,
    MIN(salary) AS minimum_salary
FROM employees
--------------
```

```
+-----------------+--------------+----------------+---------------+---------------+
| total_employees | total_salary | average_salary | maximum_salary | minimum_salary |
+-----------------+--------------+----------------+---------------+---------------+
|              10 |    630000.00 |   63000.000000 |       90000.00 |       40000.00 |
+-----------------+--------------+----------------+---------------+---------------+
```

```
-------------
SELECT
    department_id,
    COUNT(*) AS employee_count,
    SUM(salary) AS total_salary,
    AVG(salary) AS average_salary,
    MAX(salary) AS maximum_salary,
    MIN(salary) AS minimum_salary
FROM employees
GROUP BY department_id
-------------
```

department_id	employee_count	total_salary	average_salary	maximum_salary	minimum_salary
1	3	225000.00	75000.000000	90000.00	60000.00
2	2	115000.00	57500.000000	65000.00	50000.00
3	2	125000.00	62500.000000	80000.00	45000.00
4	2	125000.00	62500.000000	70000.00	55000.00
5	1	40000.00	40000.000000	40000.00	40000.00

department_id	employee_count	average_salary
1	3	75000.000000
3	2	62500.000000
4	2	62500.000000

employee_id	employee_name	salary	department_id
102	Rahul	75000.00	1
104	Priya	65000.00	2
106	Neha	80000.00	3
108	Anjali	70000.00	4
110	Pooja	90000.00	1

employee_id	employee_name	salary	department_id
103	Sneha	50000.00	2
104	Priya	65000.00	2
107	Karan	55000.00	4
108	Anjali	70000.00	4

employee_id	employee_name	salary	department_id
103	Sneha	50000.00	2
104	Priya	65000.00	2
107	Karan	55000.00	4
108	Anjali	70000.00	4

employee_id	employee_name	salary	department_id
103	Sneha	50000.00	2
104	Priya	65000.00	2

employee_id	employee_name	salary	department_id
101	Amit	60000.00	1
102	Rahul	75000.00	1
105	Rohan	45000.00	3
106	Neha	80000.00	3
110	Pooja	90000.00	1

```
-------------
SELECT
    e.employee_name,
    e.salary,
    d.department_name,
    d.location
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE e.salary > (
    SELECT AVG(salary)
    FROM employees
)
-------------
```

employee_name	salary	department_name	location
Rahul	75000.00	IT	Pune
Priya	65000.00	HR	New York
Neha	80000.00	Finance	Mumbai
Anjali	70000.00	Sales	New York
Pooja	90000.00	IT	Pune